

DEVAstra — VERIFIED WRITTEN RESOURCES (GFG VS W3SCHOOLS)

Distinct, Unique URLs per Tier • 18 Domains • Beginner / Intermediate / Advanced

This curriculum strictly evaluates **GeeksforGeeks (GFG)** and **W3Schools**. Every single entry has been verified with **unique, exact URL paths** to distinguish Beginner, Intermediate, and Advanced sub-topics perfectly without any duplication.

1. DJANGO			BACKEND PYTHON WEB FRAMEWORK
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK	
BEGINNER	W3Schools Django Introduction & Basic Syntax W3Schools Django Track	https://www.w3schools.com/django/	
INTERMEDIATE	GFG Django Tutorial - Models, Views & URL Routing GeeksforGeeks Django Hub	https://www.geeksforgeeks.org/django-tutorial/	
ADVANCED	GFG Building REST APIs with Django REST Framework (DRF) GeeksforGeeks DRF	https://www.geeksforgeeks.org/python/how-to-create-a-basic-api-using-django-rest-framework/	

2. SQL & RELATIONAL DATABASES			DATABASE QUERYING & ADMINISTRATION
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK	
BEGINNER	W3Schools SQL Tutorial - Interactive Querying & CRUD Commands W3Schools SQL	https://www.w3schools.com/sql/	
INTERMEDIATE	W3Schools SQL Tutorial - Joins, Group By, Subqueries W3Schools SQL Joins	https://www.w3schools.com/sql/sql_join.asp	
ADVANCED	GFG Advanced SQL: Window Functions & Indexing GeeksforGeeks Advanced SQL	https://www.geeksforgeeks.org/sql/sql-advanced-functions/	

3. MACHINE LEARNING (ML)			STATISTICAL AI & PREDICTIVE ALGORITHMS
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK	
BEGINNER	W3Schools Machine Learning Tutorial - Practical Python ML W3Schools Machine Learning	https://www.w3schools.com/python/python_ml_getting_started.asp	
INTERMEDIATE	GFG Supervised & Unsupervised Learning Algorithms GeeksforGeeks ML Core	https://www.geeksforgeeks.org/machine-learning/	
ADVANCED	GFG Deep Learning & Neural Networks Architecture GeeksforGeeks Deep Learning	https://www.geeksforgeeks.org/deep-learning/deep-learning-101/	

4. DATA ANALYSIS		PANDAS, NUMPY & EXPLORATORY ANALYTICS
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools Pandas Tutorial - DataFrames & CSV Parsing W3Schools Pandas	https://www.w3schools.com/python/pandas/
INTERMEDIATE	GFG Data Analysis with Python - Merging & Cleaning GeeksforGeeks Data Analysis	https://www.geeksforgeeks.org/data-analysis-with-python-tutorial/
ADVANCED	W3Schools Data Visualization with Matplotlib W3Schools Matplotlib	https://www.w3schools.com/python/matplotlib_intro.asp

5. HTML & CSS		WEB MARKUP & LAYOUT ARCHITECTURE
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools HTML5 Tutorial - Semantic Tags, Forms & Elements W3Schools HTML	https://www.w3schools.com/html/
INTERMEDIATE	W3Schools CSS Flexbox & Responsive Design W3Schools CSS Flexbox	https://www.w3schools.com/css/css3_flexbox.asp
ADVANCED	W3Schools CSS3 Animations & Keyframes W3Schools Advanced CSS	https://www.w3schools.com/css/css3_animations.asp

6. NODE.JS		ASYNCHRONOUS BACKEND JAVASCRIPT RUNTIME
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools Node.js Tutorial - Modules & HTTP Servers W3Schools Node.js	https://www.w3schools.com/nodejs/
INTERMEDIATE	W3Schools Node.js MySQL Database Integration W3Schools Node.js MySQL	https://www.w3schools.com/nodejs/nodejs_mysql.asp
ADVANCED	GFG Node.js Event Loop & Multithreading GeeksforGeeks Advanced Node	https://www.geeksforgeeks.org/node-js-event-loop/

7. GIT & GITHUB		DISTRIBUTED VERSION CONTROL & COLLABORATION
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools Git Tutorial - init, commit, push, pull W3Schools Git	https://www.w3schools.com/git/
INTERMEDIATE	W3Schools Git Branching, Merging & Conflict Fixes W3Schools Git Branching	https://www.w3schools.com/git/git_branching.asp
ADVANCED	GFG Git Internals, Reflog & Rebase Workflows GeeksforGeeks Advanced Git	https://www.geeksforgeeks.org/git-rebase/

8. DOCKER			CONTAINER PACKAGING & INFRASTRUCTURE ISOLATION
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK	
BEGINNER	GFG Docker Tutorial for Beginners - Containers & Images GeeksforGeeks Docker Hub	https://www.geeksforgeeks.org/docker-tutorial/	
INTERMEDIATE	GFG Docker Compose & Multi-Container Setup GeeksforGeeks Docker Compose	https://www.geeksforgeeks.org/docker-compose-tutorial/	
ADVANCED	GFG Docker Architecture & Networking GeeksforGeeks Docker Architecture	https://www.geeksforgeeks.org/docker-architecture/	

9. CLOUD COMPUTING			DISTRIBUTED ARCHITECTURE & CLOUD PROVIDERS
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK	
BEGINNER	GFG Introduction to Cloud Computing & Service Models GeeksforGeeks Cloud Core	https://www.geeksforgeeks.org/cloud-computing/	
INTERMEDIATE	GFG AWS Tutorial - Core Services (EC2, S3, IAM, VPC) GeeksforGeeks AWS Hub	https://www.geeksforgeeks.org/aws-tutorial/	
ADVANCED	GFG Cloud Architecture & Scalability GeeksforGeeks Cloud Architecture	https://www.geeksforgeeks.org/cloud-computing-architecture/	

10. CYBERSECURITY			INFORMATION SECURITY & THREAT ANALYSIS
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK	
BEGINNER	W3Schools Cybersecurity Tutorial - Threats & Malware W3Schools Cyber Security	https://www.w3schools.com/cybersecurity/	
INTERMEDIATE	GFG Cyber Security Tutorial - Firewalls & AppSec GeeksforGeeks CyberSec Hub	https://www.geeksforgeeks.org/cyber-security-tutorial/	
ADVANCED	GFG Cryptography & Network Security Principles GeeksforGeeks Cryptography	https://www.geeksforgeeks.org/cryptography-and-network-security/	

11. UI/UX DESIGN			DESIGN SYSTEMS, WIREFRAMING & USABILITY
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK	
BEGINNER	GFG UI/UX Design Tutorial & Principles GeeksforGeeks UI/UX	https://www.geeksforgeeks.org/ui-ux-design-tutorial/	
INTERMEDIATE	GFG Figma Tutorial - Auto Layout & Prototyping GeeksforGeeks Figma	https://www.geeksforgeeks.org/figma-tutorial/	
ADVANCED	GFG User Experience (UX) Design & Architecture GeeksforGeeks UX Design	https://www.geeksforgeeks.org/user-experience-ux-design/	

12. JAVA		ENTERPRISE OBJECT-ORIENTED PROGRAMMING
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools Java Tutorial - Syntax, OOP, Classes & Methods W3Schools Java	https://www.w3schools.com/java/
INTERMEDIATE	GFG Java Collections Framework Tutorial GeeksforGeeks Java Collections	https://www.geeksforgeeks.org/java-collection-tutorial/
ADVANCED	GFG Multithreading & Concurrency in Java GeeksforGeeks Java Advanced	https://www.geeksforgeeks.org/multithreading-in-java/

13. C++		HIGH-PERFORMANCE SYSTEMS & COMPETITIVE PROGRAMMING
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools C++ Tutorial - Variables, Loops, Functions W3Schools C++	https://www.w3schools.com/cpp/
INTERMEDIATE	GFG C++ Standard Template Library (STL) GeeksforGeeks C++ STL	https://www.geeksforgeeks.org/cpp-stl-tutorial/
ADVANCED	GFG Smart Pointers & Memory Management in C++ GeeksforGeeks Advanced C++	https://www.geeksforgeeks.org/smart-pointers-cpp/

14. GOLANG (GO)		CONCURRENT CLOUD RUNTIMES & MICROSERVICES
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools Go Tutorial - Structs, Arrays, Slices W3Schools Go	https://www.w3schools.com/go/
INTERMEDIATE	GFG Golang Tutorial - Complete Guide GeeksforGeeks Go	https://www.geeksforgeeks.org/golang/
ADVANCED	GFG Concurrency in Golang (Goroutines & Channels) GeeksforGeeks Advanced Go	https://www.geeksforgeeks.org/concurrency-in-golang/

15. KUBERNETES (K8S)		PRODUCTION CONTAINER ORCHESTRATION
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	GFG Kubernetes Tutorial - Master/Worker Nodes GeeksforGeeks K8s	https://www.geeksforgeeks.org/kubernetes-tutorial/
INTERMEDIATE	GFG Kubernetes Pods & Services GeeksforGeeks K8s Pods	https://www.geeksforgeeks.org/kubernetes-pods/
ADVANCED	GFG Kubernetes Architecture & Components GeeksforGeeks K8s Architecture	https://www.geeksforgeeks.org/kubernetes-architecture/

16. PYTHON		GENERAL PURPOSE & SCIENTIFIC COMPUTING
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools Python Tutorial - Syntax, Loops & File I/O W3Schools Python	https://www.w3schools.com/python/
INTERMEDIATE	GFG Python OOPs Concepts & Decorators GeeksforGeeks Python OOP	https://www.geeksforgeeks.org/python-oops-concepts/
ADVANCED	GFG Python Multithreading & Concurrency GeeksforGeeks Python Advanced	https://www.geeksforgeeks.org/python-multithreading/

17. REACT		DECLARATIVE COMPONENT UI FRAMEWORK
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools React Tutorial - JSX, Components W3Schools React	https://www.w3schools.com/react/
INTERMEDIATE	W3Schools React Hooks (useEffect, useState, useContext) W3Schools React Hooks	https://www.w3schools.com/react/react_hooks.asp
ADVANCED	GFG Redux Toolkit & State Management in React GeeksforGeeks React Redux	https://www.geeksforgeeks.org/redux-tutorial/

18. JAVASCRIPT		DYNAMIC CLIENT & WEB SCRIPTING
LEVEL	CURATED RESOURCE & SOURCE (GFG / W3S)	DIRECT WRITTEN TUTORIAL LINK
BEGINNER	W3Schools JavaScript Tutorial - Syntax & DOM Manipulation W3Schools JS	https://www.w3schools.com/js/
INTERMEDIATE	W3Schools JavaScript Asynchronous (Callbacks & Promises) W3Schools JS Async	https://www.w3schools.com/js/js_async.asp
ADVANCED	GFG JS Event Loop, Microtask Queue & Prototypes GeeksforGeeks JS Event Loop	https://www.geeksforgeeks.org/javascript-event-loop/